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# Oral health assessment of older adults living in Long-Term Care Facilities by non-dental professionals: A systematic review protocol of instruments



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# TITLE PAGE

Oral health assessment of older adults living in Long-Term Care Facilities by non-dental professionals: A systematic review protocol of instruments

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## ABSTRACT

**Introduction** Regular oral health assessment amongst older adults living in Long-term Care Facilities (LTCF) can improve their oral health. Different instruments have been developed and used to evaluate the oral health of institutionalized older people by non-dental professionals. These instruments must demonstrate adequate measurement properties. This systematic review aims to examine the studies describing the instruments employed to assess the oral health of older adults living in LTCF by non-dental professionals. The study will also evaluate the measurement properties of such instruments using the checklist proposed by the Consensus-based Standards to select health Measurement Instruments (COSMIN).

**Methods and analysis:** Studies describing the development of instruments for assessing oral health of institutionalized older adults by non-dental professionals will be included. Studies assessing at least one measurement property (validity, reliability, or responsiveness) will be also considered. Electronic searches will be conducted on MEDLINE (PubMed, Ovid), Embase, Web of Science, Scopus, and LILACS databases. Two independent reviewers will select the studies and will extract data concerning the characteristics of the research and the instrument. The measurement properties will be evaluated using the COSMIN checklist. The Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach will be used to grade the quality (or certainty) of evidence and strength of recommendations.

**Ethics and dissemination:** No ethical approval is required. The results will be submitted for publication to a peer review journal and presented at relevant conferences.

**Systematic review registration:** PROSPERO CRD42020191479


**Keywords:** Oral health. Long term care. Older people. Psychometrics. Geriatric Assessment

**Article Summary**

**Strengths and limitations of this study**

- This study will be an evidence for implementing best oral health care practices in Long-Term Care Facilities;

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- The systematic review will be guided by validated and consolidated instruments and tools. The assessment of the risk of bias and the quality of the evidence will be carried out using COSMIN checklist, and GRADE approach and will be reported according to the preferred reporting items for systematic reviews;

- The systematic review will allow us to support an implementation study for oral care;

- This review will be limited to studies published in English, Spanish, and Portuguese languages.

## INTRODUCTION

Population aging is a worldwide phenomenon resulting in a burden of chronic age-related conditions that constitute a significant challenge for health systems. [1] Since frailty and functional dependency increase with aging, it can be expected an increase of institutionalization among older people. [2] Institutionalized older adults present greater dependency to perform routine activities on daily basis than non-institutionalized ones, including dependency to perform oral hygiene. [3] The decline on manual dexterity and vision impairment may affect their ability to maintain an adequate level of oral hygiene and, cognitive and physical disabilities may have a negative impact on the motivation for oral health self-care. [4] Over the recent years, the complex interrelationships between frailty, cognitive decline and poor self-care related to oral health in older adults have been drawn attention due to the increase of tooth retention in this age group. The latter has been a common finding in dental surveys in different countries. The longer tooth retention may be associated with a higher prevalence of oral diseases, such as tooth and root decay and periodontal disease. [4] Older adults with missing teeth might also experience oral functional impairments due to poor adaption of dental prostheses. Another problem is the paucity of public dental services for older adults in some places, where emergency dental care is the only service available [4] and there is a shortage of oral health professionals working at long-term care facilities.

The maintenance of oral health must be part of the health care in Long-Term Care Facilities (LTCF), [5] which in turn may prevent the occurrence of systemic diseases such as aspiration pneumonia. [6] Regular oral examination can detect early dental problems, reducing oral health deterioration and the need of complex dental treatments. The systematic evaluation of oral health can also support the need for adoption, reinforcement, or improvement of guidelines for oral health promotion and

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preventive strategies in institutionalized older adults. In this context, caregivers are responsible for health care of older people in the daily routine in LTCF. Thus, they have a crucial role in performing oral health evaluation as part of the health care plan. [4,5,7,8]

Some instruments have been used to evaluate the oral health of institutionalized older people by non-dental professionals, such as Oral Health Assessment Tool (OHAT), [9] Minimal Data Set (MDS), [10] Dental Hygiene Registration (DHR), [11] Oral Health Screening Tool for Nursing Personnel (OHSTNP), [12] and InterRAI. [13] The use of appropriate instruments is paramount to identify oral health problems, favoring the daily oral care plan, and referral to a dental service, if necessary**.** [7,11,13–15] Although such instruments must demonstrate adequate measurement properties, there is evidence that some of them may present methodological shortcomings (15,16). Two previous systematic reviews have evaluated the instruments used for oral health assessment [15,16]. However, both studies have not focused on institutionalized older adults [15,16]. Thus, an update seems timely and legitimate.

Implementing best practices in LTCF, such as oral health assessment, should be evidence-based and adapted to each context. This review can support the identification of the most suitable instrument for oral health evaluation of institutionalized older adults. The results may also indicate the need for additional assessment of the measurement properties of the existing instruments or highlight the need to develop and test a new one. This systematic review aims to identify the studies that described instruments developed to assess the oral health of older adults in Long-Term Care Facilities by non-dental professionals and to evaluate their measurement properties.

## METHODS AND ANALYSIS

This study protocol has been prepared according to the 2015 Prefered Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines (Supplemental material 1). The systematic review also will be reported according to the PRISMA [17] and COSMIN checklist for systematic reviews. [18] The protocol of this systematic review was registered in the PROSPERO database (Ref: CRD42020191479). If there are important changes in the study, these will be documented in the PROSPERO database.

### Review question

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What instruments have been used to assess the oral health of institutionalized older people by non-dental professionals.

Do these instruments present acceptable measurement properties?

**Patient and public involvement**

No patients were involved in the design of the study.

**Inclusion criteria**

- The instrument must aim to assess oral health of institutionalized older adults;
- The instrument must be applied by non-dental professionals, such as caregivers and nurses;
- Studies must describe the development or the assessment of at least one measurement property of the instrument (validity, reliability, or responsiveness);
- Studies published in English, Spanish or Portuguese;
- No publication date or publication status restrictions will be imposed.

**Types of studies**

This review will include epidemiologic studies, including validation studies and observational studies reporting measurement properties of the instruments used to assess oral health of institutionalized older adults by non-dental professionals.

**Search strategy**

The studies will be identified by searching four electronic databases: MEDLINE (PubMed), Embase, Web of Science, Scopus, and LILACS (Latin American and Caribbean Literature in Health Sciences). Besides, a search on Google Scholar and OpenGray will identify further potential relevant articles. The reference lists of included articles will also be screened.

According to the COSMIN checklist for systematic reviews, the search strategy will follow up four key elements of the review aim: 1. Construct (related to Oral Health); 2. Population (related to Age, Long-term care and Caregivers); 3. Type of instrument (related to Assessment Tool), and 4. Measurement properties (using a PubMed search filter to find studies on measurement properties. [19] The Search Blocks site will be used to assist the construction of the search strategy blocks

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(https://blocks.bmi-online.nl/). The index terms were used to develop a full search strategy for MEDLINE (Supplemental material 2).

Screening and eligibility assessment will be performed independently by two trained reviewers. They will evaluate the relevant articles according to abstract and then the full-text, respectively. Disagreements between the reviewers will be resolved by consensus. The number of identified articles, the screening and eligibility steps, and the included studies will be shown using the flow diagram proposed by PRISMA.

**Data extraction**

Two reviewers will perform data extraction independently using a prepiloted spreadsheet developed for this purpose. A third reviewer will be consulted in case of disagreements. The characteristics of the study and the instrument that will be extracted from each article are presented in Table 1.

Table 1: Data to be extracted from included studies

| **Characteristics of the study** |
|---|
| Aim of the study |
| Study design |
| Sample size |
| Country of data collection |
| Settings |
| Measurement properties evaluated |
| Limitations of the study |
| Conclusions |
| **Characteristics of the instrument** |
| Name of the instrument |
| Instrument objectives |
| Professionals who applied the instrument |
| Need for training to apply the instrument |
| Instrument items |
| Instrument scoring system |

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The extracted data from all included studies will be qualitatively synthesized and presented in tables along with a narrative summary.

**Assessment of methodological quality**

Two trained reviewers will independently evaluate the quality of the included studies using the COSMIN checklist. [20] The COSMIN aims to improve the selection of Patient-Reported Outcome Measures (PROMs) instruments in research and clinical practice through tools to select the most appropriate instrument. Despite focusing on PROMs, it can also be adapted for other purposes, such as instruments for measuring clinical outcomes. [18] According to COSMIN, three domains must be evaluated to assess the quality of the instrument: reliability, validity, and responsiveness. Each domain contains one or more measurement properties (Table 2). [18] Figure 1 presents the steps to assess the methodological quality and grading the quality of evidence.

Table 2: Definitions of measurement properties

| **Domain** | **Definition** |
|---|---|
| Reliability | The degree to which the measurement is free from measurement error |
| Validity | The degree to which a PROM measures the construct(s) it purports to measure |
| Responsiveness | The ability of a PROM to detect change over time in the construct to be measured |

The Risk of Bias checklist will be used to assess the methodological quality of single studies included in the systematic review. The checklist contains standards referring to design requirements and preferred statistical methods of studies on measurement properties, containing ten boxes: with one standard for PROM development and nine for measurement properties: Content validity, Structural validity, Internal consistency, Cross-cultural validity/measurement invariance, Reliability, Measurement error, Criterion validity, Hypotheses testing for construct validity, and Responsiveness. A standard COSMIN Excel spreadsheet will be used for each box. The measurement properties evaluated in a study will determine which boxes should be completed. A four-point rating scale will be assigned for each study as follows “very good”, “adequate”, “doubtful” or “inadequate”. The overall rating of the quality of each study will be determined by taking the lowest rating of any standard in the box. [18]

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Next, the results of each study on a measurement property will be rated against the updated criteria for good measurement properties. Each result will be rated as either sufficient (+), insufficient (-), or indeterminate (?). [18] The overview tables will be used to assess whether the results of two or more articles using the same instrument are consistent with each other to obtain the pooled result against criteria of good measurement properties.

**Grading the quality of evidence**

After pooling all evidence per measurement property per instrument and rating the pooled result against the criteria for good measurement properties, the quality of the evidence will be graded based on a modified Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach for systematic reviews. The quality of the evidence refers to the confidence that the summarized result is valid. The GRADE approach classifies the certainty in the findings as high, moderate, low, or very low. Four of the five factors of the GRADE approach will be used for evaluating measurement properties: risk of bias (the methodological quality of the studies), inconsistency (unexplained inconsistency of results across studies), imprecision (total sample size of available studies), and indirectness (evidence from a different population than the population of interest in the review). [21] The quality of the evidence will be performed by two trained reviewers independently. The GRADE approach will be used to downgrade evidence when there are concerns about the quality of evidence. The starting point is always the assumption that the pooled is of high quality. The quality of evidence can be subsequently downgraded by one or two levels per factor to moderate, low, or very low evidence if there is imprecision (low sample size) or serious or very serious risk of bias, inconsistency, indirectness. The quality of evidence can even be downgraded by three levels when the evidence is based on only one inadequate study (extremely serious risk of bias).

This review also aims to assess the interpretability, which corresponds to the qualitative meaning for the quantitative scores obtained by applying the instrument, and the feasibility, that is, the ease of use of the instruments. [18] Finally, based on the evaluations performed, this study will might be able to suggest recommendations on the most suitable instrument for non-dental professionals assessing the oral health of institutionalized older adults.

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## Ethics and dissemination

No ethical approval is required. The results will be submitted for publication to a scientific journal and presented at relevant conferences.

**Authors Statement:** LGR and RCF conceived the study, LGR and ILF drafted the manuscript, AAS, MVV and RCF revised the manuscript. All authors approves the final version. LGR is the guarantor of the review.

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**Competing interests statement:** None declared.

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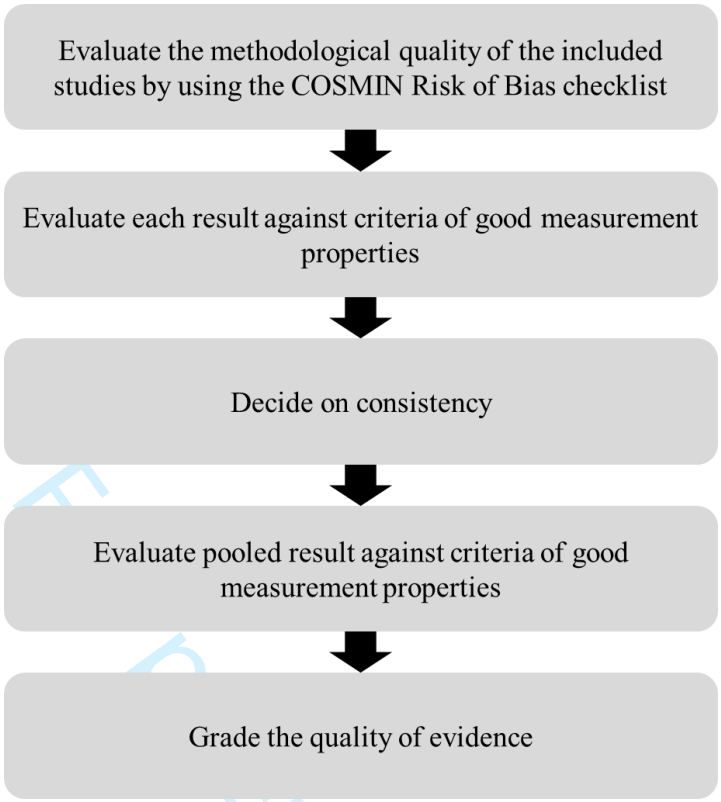
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Figure 1: Flowchart of methodological quality evaluation and grading the quality of evidence

Source: Adapted from Mokkink et al., 2018 [18]

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Supplemental material

| Search | Query | Results |
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| #1 | """Oral Health""[Mesh] OR ""Geriatric Dentistry""[Mesh] OR ""Dental Care""[Mesh] OR ""Dental Care for Aged""[Mesh] OR ""Dentistry""[Mesh] OR ""Dentists""[Mesh] OR dental*[ti] OR dentist*[ti] OR oral hygien*[ti] OR dental*[ot] OR dentist*[ot] OR oral hygien*[ot]",Publication Date,,"""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]" | 499,813 |
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| | Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]" | |
| #4 | """Oral Health screening tool"" OR ""Geriatric Assessment""[Mesh] OR ""assessment tool"" OR ""needs assessment""[Mesh] OR ""oral assessment"" OR ""assessment scale"" OR ""Surveys and Questionnaires""[Mesh] OR questionnaire*[tiab] OR instrument*[tiab] OR measure*[tiab] OR survey*[tiab]",Publication Date,,"""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]" | 4,917,678 |
| #5 | "(instrumentation[sh] OR methods[sh] OR ""Validation Studies""[pt] OR ""Comparative Study""[pt] OR ""psychometrics""[MeSH] OR psychometr*[tiab] OR clinimetr*[tw] OR clinometr*[tw] OR ""outcome assessment (health care)""[MeSH] OR ""outcome assessment""[tiab] OR ""outcome measure*""[tw] OR ""observer variation""[MeSH] OR ""observer variation""[tiab] OR ""Health Status Indicators""[Mesh] OR ""reproducibility of results""[MeSH] OR reproducib*[tiab] OR ""discriminant analysis""[MeSH] OR reliab*[tiab] OR unreliab*[tiab] OR valid*[tiab] OR ""coefficient of variation""[tiab] OR coefficient[tiab] OR homogeneity[tiab] OR homogeneous[tiab] OR ""internal consistency""[tiab] OR (cronbach*[tiab] AND (alpha[tiab] OR alphas[tiab])) OR (item[tiab] AND (correlation*[tiab] OR selection*[tiab] OR reduction*[tiab])) OR agreement[tw] OR precision[tw] OR imprecision[tw] OR ""precise values""[tw] OR test-retest[tiab] OR (test[tiab] AND retest[tiab]) OR (reliab*[tiab] AND (test[tiab] OR retest[tiab])) OR stability[tiab] OR interrater[tiab] OR inter-rater[tiab] OR intrarater[tiab] OR intra-rater[tiab] OR intertester[tiab] OR inter-tester[tiab] OR intratester[tiab] OR intra-tester[tiab] OR interobserver[tiab] OR inter-observer[tiab] OR intraobserver[tiab] OR intra-observer[tiab] OR intertechnician[tiab] OR inter-technician[tiab] OR intratechnician[tiab] OR intra-technician[tiab] OR interexaminer[tiab] OR inter-examiner[tiab] OR intraexaminer[tiab] OR intra-examiner[tiab] OR interassay[tiab] OR inter-assay[tiab] OR intraassay[tiab] OR intra-assay[tiab] OR interindividual[tiab] OR inter-individual[tiab] OR intraindividual[tiab] OR intra-individual[tiab] OR interparticipant[tiab] OR inter-participant[tiab] OR intraparticipant[tiab] OR intra-participant[tiab] OR kappa[tiab] OR kappa's[tiab] OR kappas[tiab] OR repeatab*[tw] OR ((replicab*[tw] OR repeated[tw]) AND (measure[tw] OR measures[tw] OR findings[tw] OR result[tw] OR results[tw] OR test[tw] OR tests[tw])) OR generaliza*[tiab] OR generalisa*[tiab] OR concordance[tiab] OR (intraclass[tiab] AND correlation*[tiab]) OR discriminative[tiab] OR ""known group""[tiab] OR ""factor analysis""[tiab] OR ""factor analyses""[tiab] OR ""factor structure""[tiab] OR ""factor structures""[tiab] OR dimension*[tiab] OR subscale*[tiab] OR (multitrait[tiab] AND scaling[tiab] AND (analysis[tiab] OR analyses[tiab])) OR ""item discriminant""[tiab] OR ""interscale correlation*""[tiab] OR error[tiab] OR errors[tiab] OR ""individual variability""[tiab] OR ""interval variability""[tiab] OR ""rate variability""[tiab] OR (variability[tiab] AND (analysis[tiab] OR values[tiab])) OR (uncertainty[tiab] AND (measurement[tiab] OR measuring[tiab])) OR ""standard | 9,508,385 |

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| | error of measurement""[tiab] OR sensitiv*[tiab] OR responsive*[tiab] OR (limit[tiab] AND detection[tiab]) OR ""minimal detectable concentration""[tiab] OR interpretab*[tiab] OR ((minimal[tiab] OR minimally[tiab] OR clinical[tiab] OR clinically[tiab]) AND (important[tiab] OR significant[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR (small*[tiab] AND (real[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR ""meaningful change""[tiab] OR ""ceiling effect""[tiab] OR ""floor effect""[tiab] OR ""Item response model""[tiab] OR IRT[tiab] OR Rasch[tiab] OR ""Differential item functioning""[tiab] OR DIF[tiab] OR ""computer adaptive testing""[tiab] OR ""item bank""[tiab] OR ""cross-cultural equivalence""[tiab])",Publication Date,,""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra-observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter-technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR ((""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR | |

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| | (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract]" | |
| #6 | "(""addresses""[Publication Type] OR ""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""lectures""[Publication Type] OR ""legal cases""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""popular works""[Publication Type] OR ""congresses""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms])",Publication Date,,"(""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms])" | 4,188,714 |
| #7 | #1 AND #2 AND #3 AND #4 AND #5,Publication Date,,"(""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]) AND (""Aged""[MeSH Terms] OR ""aged, 80 and over""[MeSH Terms] OR ""Frail Elderly""[MeSH Terms] OR ""Geriatrics""[MeSH Terms] OR ""Geriatric Psychiatry""[MeSH Terms] OR ""Geriatric Nursing""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Health Services for the Aged""[MeSH Terms] OR (""elder*""[Text Word] OR ""eldest""[Text Word] | 423 |

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| | OR ""frail*""[Text Word] OR ""geriatri*""[Text Word] OR ""old age*""[Text Word] OR ""oldest old*""[Text Word] OR ""senior*""[Text Word] OR ""senium""[Text Word] OR ""very old*""[Text Word] OR ""septuagenarian*""[Text Word] OR ""octagenarian*""[Text Word] OR ""octogenarian*""[Text Word] OR ""nonagenarian*""[Text Word] OR ""centarian*""[Text Word] OR ""centenarian*""[Text Word] OR ""supercentenarian*""[Text Word] OR ""older people""[Text Word] OR ""older subject*""[Text Word] OR ""older patient*""[Text Word] OR ""older age*""[Text Word] OR ""older adult*""[Text Word] OR ""older man""[Text Word] OR ""older men""[Text Word] OR ""older male*""[Text Word] OR ""older woman""[Text Word] OR ""older women""[Text Word] OR ""older female*""[Text Word] OR ""older population*""[Text Word] OR ""older person*""[Text Word])) AND (""nursing homes""[MeSH Terms] OR ""long term care""[MeSH Terms] OR ""residential care""[All Fields] OR ""homes for the aged""[MeSH Terms] OR ""residential facilities""[MeSH Terms] OR ""long-term care institution""[All Fields] OR ""residential aged care""[All Fields] OR ""skilled nursing facilities""[MeSH Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]) AND (""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]) AND (""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra- | |

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| | observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter-technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR (""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract])" | |
| #8 | #7 NOT #6,Publication Date,,"((""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]) AND (""Aged""[MeSH Terms] OR ""aged, 80 and over""[MeSH Terms] OR ""Frail Elderly""[MeSH Terms] OR ""Geriatrics""[MeSH Terms] OR ""Geriatric Psychiatry""[MeSH Terms] OR ""Geriatric Nursing""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Health Services for the Aged""[MeSH Terms] OR (""elder*""[Text Word] OR ""eldest""[Text Word] OR ""frail*""[Text Word] OR ""geriatri*""[Text Word] OR ""old age*""[Text | 420 |

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| | Word] OR ""oldest old*""[Text Word] OR ""senior*""[Text Word] OR ""senium""[Text Word] OR ""very old*""[Text Word] OR ""septuagenarian*""[Text Word] OR ""octagenarian*""[Text Word] OR ""octogenarian*""[Text Word] OR ""nonagenarian*""[Text Word] OR ""centarian*""[Text Word] OR ""centenarian*""[Text Word] OR ""supercentenarian*""[Text Word] OR ""older people""[Text Word] OR ""older subject*""[Text Word] OR ""older patient*""[Text Word] OR ""older age*""[Text Word] OR ""older adult*""[Text Word] OR ""older man""[Text Word] OR ""older men""[Text Word] OR ""older male*""[Text Word] OR ""older woman""[Text Word] OR ""older women""[Text Word] OR ""older female*""[Text Word] OR ""older population*""[Text Word] OR ""older person*""[Text Word])) AND (""nursing homes""[MeSH Terms] OR ""long term care""[MeSH Terms] OR ""residential care""[All Fields] OR ""homes for the aged""[MeSH Terms] OR ""residential facilities""[MeSH Terms] OR ""long-term care institution""[All Fields] OR ""residential aged care""[All Fields] OR ""skilled nursing facilities""[MeSH Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]) AND (""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]) AND (""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra-observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter- | |

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| | technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR (""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract])) NOT ((""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms]))" | |

**PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol***

| Section and topic | Item No | Checklist item | Page Number |
|---|---|---|---|
| **ADMINISTRATIVE INFORMATION** | | | |
| Title: | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | 1 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | NA |
| Registration | 2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | 2,5 |
| Authors: | | | |
| Contact | 3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | 1 |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | 8 |
| Amendments | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | 4 |
| Support: | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | 9 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | NA |
| Role of sponsor or funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | NA |
| **INTRODUCTION** | | | |
| Rationale | 6 | Describe the rationale for the review in the context of what is already known | 4 |
| Objectives | 7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | 4 |
| **METHODS** | | | |
| Eligibility criteria | 8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | 5 |
| Information sources | 9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | 5 |
| Search strategy | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be | Supplemental |

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| | | | |
|---|---|---|---|
| | | repeated | material |
| Study records: | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | 5,6 |
| Selection process | 11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | 5,6 |
| Data collection process | 11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | 6 |
| Data items | 12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | 6 |
| Outcomes and prioritization | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | 6 |
| Risk of bias in individual studies | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | 6-8 |
| Data synthesis | 15a | Describe criteria under which study data will be quantitatively synthesised | 7,8 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as $I^2$, Kendall's $\tau$) | 8 |
| | 15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | 8 |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | 8 |
| Meta-bias(es) | 16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | 8 |
| Confidence in cumulative evidence | 17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | 8 |

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

*From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.*

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# Oral health assessment of older adults living in Long-Term Care Facilities by non-dental professionals: A systematic review protocol of instruments



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# TITLE PAGE

Oral health assessment of older adults living in Long-Term Care Facilities by non-dental professionals: A systematic review protocol of instruments

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## ABSTRACT

**Introduction** Regular oral health assessment amongst older adults living in Long-term Care Facilities (LTCF) can improve their oral health. Different instruments have been developed and used to evaluate the oral health of institutionalized older people by non-dental professionals. These instruments must demonstrate adequate measurement properties. This systematic review aims to examine the studies describing the instruments employed to assess the oral health of older adults living in LTCF by non-dental professionals. The study will also evaluate the measurement properties of such instruments using the checklist proposed by the Consensus-based Standards to select health Measurement Instruments (COSMIN).

**Methods and analysis:** Studies describing the development of instruments for assessing oral health of institutionalized older adults by non-dental professionals will be included. Studies assessing at least one measurement property (validity, reliability, or responsiveness) will be also considered. Electronic searches will be conducted on MEDLINE (PubMed, Ovid), Embase, Web of Science, Scopus, and LILACS databases. Two independent reviewers will select the studies and will extract data concerning the characteristics of the research and the instrument. The measurement properties will be evaluated using the COSMIN checklist. The Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach will be used to grade the quality (or certainty) of evidence and strength of recommendations.

**Ethics and dissemination:** No ethical approval is required. The results will be submitted for publication to a peer review journal and presented at relevant conferences.

**Systematic review registration:** PROSPERO CRD42020191479


**Keywords:** Oral health. Long term care. Older people. Psychometrics. Geriatric Assessment

**Article Summary**

**Strengths and limitations of this study**

- This study will be an evidence for implementing best oral health care practices in Long-Term Care Facilities;

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- The systematic review will be guided by validated and consolidated instruments and tools. The assessment of the risk of bias and the quality of the evidence will be carried out using COSMIN checklist, and GRADE approach and will be reported according to the preferred reporting items for systematic reviews;

- The systematic review will allow us to support an implementation study for oral care;

- This review will be limited to studies published in English, Spanish, and Portuguese languages.

## INTRODUCTION

Population aging is a worldwide process resulting in a burden of chronic age-related conditions that constitute a significant challenge for health systems. [1] Since frailty and functional dependency increase with aging, it can be expected an increase of institutionalization among older people. [2] Institutionalized older adults present greater dependency to perform routine activities on daily basis than non-institutionalized ones, including dependency to perform oral hygiene. [3] The decline on manual dexterity and vision impairment may affect their ability to maintain an adequate level of oral hygiene and, cognitive and physical disabilities may have a negative impact on the motivation for oral health self-care. [4] Over the recent years, the complex interrelationships between frailty, cognitive decline and poor self-care related to oral health in older adults have been drawn attention due to the increase of tooth retention in this age group. The latter has been a common finding in dental surveys in different countries. The longer tooth retention may be associated with a higher prevalence of oral diseases, such as tooth and root decay and periodontal disease. [4] Older adults with missing teeth might also experience oral functional impairments due to poor adaptation of dental prostheses. Another problem is the paucity of public dental services for older adults in some places, where emergency dental care is the only service available [4] and there is a shortage of oral health professionals working at long-term care facilities.

The maintenance of oral health must be part of the health care in Long-Term Care Facilities (LTCF), [5] which in turn may prevent the occurrence of systemic diseases such as aspiration pneumonia. [6] Regular oral examination can detect early dental problems, reducing oral health deterioration and the need of complex dental treatments. The systematic evaluation of oral health can also support the need for adoption, reinforcement, or improvement of guidelines for oral health promotion and preventive strategies in

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institutionalized older adults. In this context, caregivers are responsible for health care of older people in the daily routine in LTCF. Thus, they have a crucial role in performing oral health evaluation as part of the health care plan. [4,5,7,8]

Some instruments have been used to evaluate the oral health of institutionalized older people by non-dental professionals, such as Oral Health Assessment Tool (OHAT), [9] Minimal Data Set (MDS), [10] Dental Hygiene Registration (DHR), [11] Oral Health Screening Tool for Nursing Personnel (OHSTNP), [12] and oral health-related section of the Resident Assessment Instrument (ohr-InterRAI). [13] The use of appropriate instruments is paramount to identify oral health problems, favoring the daily oral care plan, and referral to a dental service, if necessary**.** [7,11,13–15] Although such instruments must demonstrate adequate measurement properties, there is evidence that some of them may present methodological shortcomings (15,16). Two previous systematic reviews have evaluated the instruments used for oral health assessment [15,16]. However, both studies have not focused on institutionalized older adults [15,16]. Thus, an update seems timely and legitimate.

Implementing best practices in LTCF, such as oral health assessment, should be evidence-based and adapted to each context. This review can support the identification of the most suitable instrument for oral health evaluation of institutionalized older adults. The results may also indicate the need for additional assessment of the measurement properties of the existing instruments or highlight the need to develop and test a new one. This systematic review aims to identify the studies that described instruments developed to assess the oral health of older adults in Long-Term Care Facilities by non-dental professionals and to evaluate their measurement properties.

## METHODS AND ANALYSIS

This study protocol has been prepared according to the 2015 Prefered Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) guidelines (Supplemental material 1). The systematic review also will be reported according to the PRISMA [17] and COSMIN checklist for systematic reviews. [18] The protocol of this systematic review was registered in the PROSPERO database (Ref: CRD42020191479). If there are important changes in the study, these will be documented in the PROSPERO database.

### Review question

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What instruments have been used to assess the oral health of institutionalized older people by non-dental professionals?

Do these instruments present acceptable measurement properties?

**Patient and public involvement**

No patients were involved in the design of the study.

**Inclusion criteria**

- The instrument must aim to assess oral health of institutionalized older adults;
- The instrument must be applied by non-dental professionals, such as caregivers and nurses;
- Studies must describe the development or the assessment of at least one measurement property of the instrument (validity, reliability, or responsiveness);
- Studies published in English, Spanish or Portuguese;
- No publication date or publication status restrictions will be imposed.

**Types of studies**

This review will include epidemiologic studies, including validation studies and observational studies reporting measurement properties of the instruments used to assess oral health of institutionalized older adults by non-dental professionals.

**Search strategy**

The studies will be identified by searching four electronic databases: MEDLINE (PubMed), Embase, Web of Science, Scopus, and LILACS (Latin American and Caribbean Literature in Health Sciences). Besides, a search on Google Scholar and OpenGray will identify further potential relevant articles. The reference lists of included articles will also be screened.

According to the COSMIN checklist for systematic reviews, the search strategy will follow up four key elements of the review aim: 1. Construct (related to Oral Health); 2. Population (related to Age, Long-term care and Caregivers); 3. Type of instrument (related to Assessment Tool), and 4. Measurement properties (using a PubMed search filter to find studies on measurement properties. [19] The Search Blocks site will be used to assist the construction of the search strategy blocks (https://blocks.bmi-online.nl/). The

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index terms were used to develop a full search strategy for MEDLINE (Supplemental material 2).

Screening and eligibility assessment will be performed independently by two trained reviewers. They will evaluate the relevant articles according to abstract and then the full-text, respectively. Disagreements between the reviewers will be resolved by consensus. The number of identified articles, the screening and eligibility steps, and the included studies will be shown using the flow diagram proposed by PRISMA.

**Data extraction**

Two reviewers will perform data extraction independently using a prepiloted spreadsheet developed for this purpose. A third reviewer will be consulted in case of disagreements. The characteristics of the study and the instrument that will be extracted from each article are presented in Table 1.

Table 1: Data to be extracted from included studies

| **Characteristics of the study** |
|---|
| Aim of the study |
| Study design |
| Sample size |
| Country of data collection |
| Settings |
| Measurement properties evaluated |
| Limitations of the study |
| Conclusions |
| **Characteristics of the instrument** |
| Name of the instrument |
| Instrument objectives |
| Professionals who applied the instrument |
| Need for training to apply the instrument |
| Instrument items |
| Instrument scoring system |

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The extracted data from all included studies will be qualitatively synthesized and presented in tables along with a narrative summary.

**Assessment of methodological quality**

Two trained reviewers will independently evaluate the quality of the included studies using the COSMIN Risk of Bias checklist. [20] The COSMIN aims to improve the selection of Patient-Reported Outcome Measures (PROMs) instruments in research and clinical practice through tools to select the most appropriate instrument. Despite focusing on PROMs, it can also be adapted for other purposes, such as instruments for measuring clinical outcomes. [18] According to COSMIN, three domains must be evaluated to assess the quality of the instrument: reliability, validity, and responsiveness. Each domain contains one or more measurement properties (Table 2). [18] Figure 1 presents the steps to assess the methodological quality and grading the quality of evidence.

Table 2: Definitions of measurement properties

| **Domain** | **Definition** |
|---|---|
| Reliability | The degree to which the measurement is free from measurement error |
| Validity | The degree to which a PROM measures the construct(s) it purports to measure |
| Responsiveness | The ability of a PROM to detect change over time in the construct to be measured |

The COSMIN Risk of Bias checklist will be used to assess the methodological quality of single studies included in the systematic review. The checklist contains standards referring to design requirements and preferred statistical methods of studies on measurement properties, containing ten boxes: with one standard for PROM development and nine for measurement properties: Content validity, Structural validity, Internal consistency, Cross-cultural validity/measurement invariance, Reliability, Measurement error, Criterion validity, Hypotheses testing for construct validity, and Responsiveness. A standard COSMIN Excel spreadsheet will be used for each box. The measurement properties evaluated in a study will determine which boxes should be completed. A four-point rating scale will be assigned for each study as follows “very good”, “adequate”,

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"doubtful" or "inadequate". The overall rating of the quality of each study will be determined by taking the lowest rating of any standard in the box. [18] Next, the results of each study on a measurement property will be rated against the updated criteria for good measurement properties. Each result will be rated as either sufficient (+), insufficient (-), or indeterminate (?). [18] The overview tables will be used to assess whether the results of two or more articles using the same instrument are consistent with each other to obtain the pooled result against criteria of good measurement properties.

**Grading the quality of evidence**

After pooling all evidence per measurement property per instrument and rating the pooled result against the criteria for good measurement properties, the quality of the evidence will be graded based on a modified Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach for systematic reviews. The quality of the evidence refers to the confidence that the summarized result is valid. The GRADE approach classifies the certainty in the findings as high, moderate, low, or very low. Four of the five factors of the GRADE approach will be used for evaluating measurement properties: risk of bias (the methodological quality of the studies), inconsistency (unexplained inconsistency of results across studies), imprecision (total sample size of available studies), and indirectness (evidence from a different population than the population of interest in the review). [21] The quality of the evidence will be performed by two trained reviewers independently. The GRADE approach will be used to downgrade evidence when there are concerns about the quality of evidence. The starting point is always the assumption that the obtained pooled result if of "High" quality level. The quality of evidence can be subsequently downgraded by one or two levels per factor to moderate, low, or very low evidence if there is imprecision (low sample size) or serious or very serious risk of bias, inconsistency, indirectness. The quality of evidence can even be downgraded by three levels when the evidence is based on only one inadequate study (extremely serious risk of bias).

This review also aims to assess the interpretability, which corresponds to the qualitative meaning for the quantitative scores obtained by applying the instrument, and the feasibility, that is, the ease of use of the instruments. [18] Finally, based on the evaluations performed, this study will might be able to suggest recommendations on the most suitable instrument for non-dental professionals assessing the oral health of institutionalized older adults.

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## Ethics and dissemination

No ethical approval is required. The results will be submitted for publication to a scientific journal and presented at relevant conferences.

**Authors Statement:** LGR and RCF conceived the study, LGR, and ILF drafted the manuscript, AAS, MVV and RCF revised the manuscript. All authors approved the final version to be published. LGR is the guarantor of the review.

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**Competing interests statement:** None declared.

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20 Mokkink LB, de Vet HCM, Prinsen CAC, et al. COSMIN Risk of Bias checklist for systematic reviews of Patient-Reported Outcomes Measures. *Qual Life Res* 2018 May;27(5):1171–9. doi:10.1007/s11136-017-1765-4.

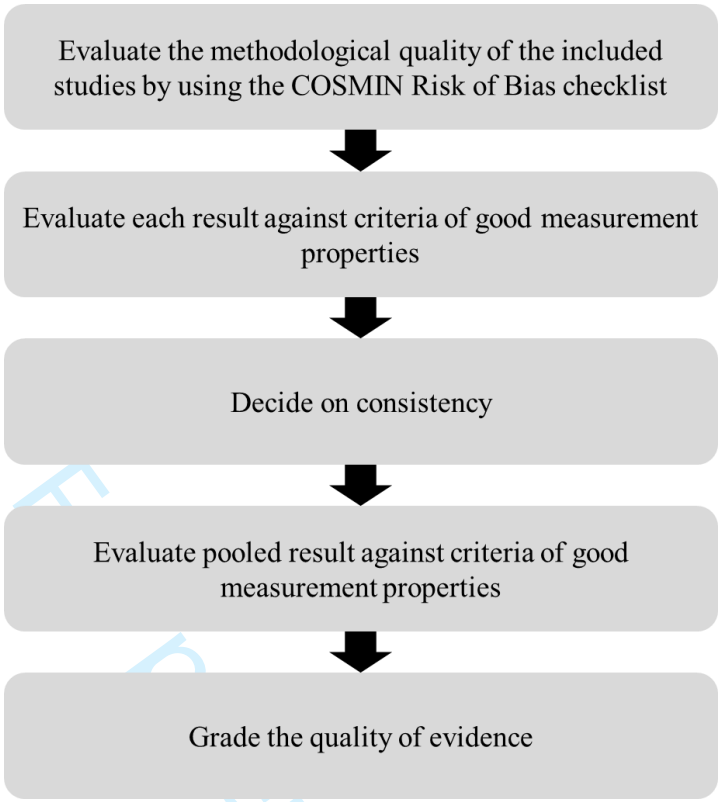
21 Schünemann HJ, Brozek J, Guyatt G, et al. GRADE Handbook - Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach. 2013. https://gdt.gradepro.org/app/handbook/handbook.html (accessed 27 Nov 2020).

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Figure 1: Flowchart of methodological quality evaluation and grading the quality of evidence

Source: Adapted from Mokkink et al., 2018 [18]

For peer review only




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Supplemental material

| Search | Query | Results |
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| #1 | """Oral Health""[Mesh] OR ""Geriatric Dentistry""[Mesh] OR ""Dental Care""[Mesh] OR ""Dental Care for Aged""[Mesh] OR ""Dentistry""[Mesh] OR ""Dentists""[Mesh] OR dental*[ti] OR dentist*[ti] OR oral hygien*[ti] OR dental*[ot] OR dentist*[ot] OR oral hygien*[ot]",Publication Date,,"""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]" | 499,813 |
| #2 | "(""Aged""[Mesh] OR ""Aged, 80 and over""[Mesh] OR ""Frail Elderly""[Mesh] OR ""Geriatrics""[Mesh] OR ""Geriatric Psychiatry""[Mesh] OR ""Geriatric Nursing""[Mesh] OR ""Geriatric Dentistry""[Mesh] OR ""Dental Care for Aged""[Mesh] OR ""Health Services for the Aged""[Mesh]) OR (elder*[tw] OR eldest[tw] OR frail*[tw] OR geriatri*[tw] OR old age*[tw] OR oldest old*[tw] OR senior*[tw] OR senium[tw] OR very old*[tw] OR septuagenarian*[tw] OR octagenarian*[tw] OR octogenarian*[tw] OR nonagenarian*[tw] OR centarian*[tw] OR centenarian*[tw] OR supercentenarian*[tw] OR older people[tw] OR older subject*[tw] OR older patient*[tw] OR older age*[tw] OR older adult*[tw] OR older man[tw] OR older men[tw] OR older male*[tw] OR older woman[tw] OR older women[tw] OR older female*[tw] OR older population*[tw] OR older person*[tw])",Publication Date,,"""Aged""[MeSH Terms] OR ""aged, 80 and over""[MeSH Terms] OR ""Frail Elderly""[MeSH Terms] OR ""Geriatrics""[MeSH Terms] OR ""Geriatric Psychiatry""[MeSH Terms] OR ""Geriatric Nursing""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Health Services for the Aged""[MeSH Terms] OR ""elder*""[Text Word] OR ""eldest""[Text Word] OR ""frail*""[Text Word] OR ""geriatri*""[Text Word] OR ""old age*""[Text Word] OR ""oldest old*""[Text Word] OR ""senior*""[Text Word] OR ""senium""[Text Word] OR ""very old*""[Text Word] OR ""septuagenarian*""[Text Word] OR ""octagenarian*""[Text Word] OR ""octogenarian*""[Text Word] OR ""nonagenarian*""[Text Word] OR ""centarian*""[Text Word] OR ""centenarian*""[Text Word] OR ""supercentenarian*""[Text Word] OR ""older people""[Text Word] OR ""older subject*""[Text Word] OR ""older patient*""[Text Word] OR ""older age*""[Text Word] OR ""older adult*""[Text Word] OR ""older man""[Text Word] OR ""older men""[Text Word] OR ""older male*""[Text Word] OR ""older woman""[Text Word] OR ""older women""[Text Word] OR ""older female*""[Text Word] OR ""older population*""[Text Word] OR ""older person*""[Text Word]" | 3,421,849 |
| #3 | """nursing homes""[Mesh] OR ""long term care""[Mesh] OR ""residential care"" OR ""homes for the aged""[Mesh] OR ""residential facilities""[Mesh] OR ""long-term care institution"" OR ""residential aged care"" OR ""skilled nursing facilities""[Mesh] OR ""nursing facility"" OR ""care worker"" OR ""nurses aids"" OR ""nursing personnel"" OR ""Caregivers""[Mesh] OR ""Nursing""[Mesh] OR ""nursing"" [Subheading] OR ""Nursing Care""[Mesh] OR ""Nurses""[Mesh] OR ""Nursing Staff""[Mesh] OR ""Nursing Care""[Mesh] OR ""Nurse's Role""[Mesh] OR ""Home Health Aides""[Mesh] OR ""Nursing Assistants""[Mesh] OR ""Licensed Practical Nurses""[Mesh] OR ""Home Care Services""[Mesh] OR ""nurs*""[tiab] OR ""home care""[tiab] OR ""home healthcare""[tiab] OR ""home health care""[tiab] OR ""care aide*""[tiab] OR ""home health aide*""[tiab]",Publication Date,,"""nursing homes""[MeSH Terms] OR ""long term care""[MeSH Terms] OR ""residential care""[All Fields] OR ""homes for the aged""[MeSH Terms] OR ""residential facilities""[MeSH Terms] OR ""long-term care institution""[All Fields] OR ""residential aged care""[All Fields] OR ""skilled nursing facilities""[MeSH | 802,910 |

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| | Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]" | |
| #4 | """Oral Health screening tool"" OR ""Geriatric Assessment""[Mesh] OR ""assessment tool"" OR ""needs assessment""[Mesh] OR ""oral assessment"" OR ""assessment scale"" OR ""Surveys and Questionnaires""[Mesh] OR questionnaire*[tiab] OR instrument*[tiab] OR measure*[tiab] OR survey*[tiab]",Publication Date,,"""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]" | 4,917,678 |
| #5 | "(instrumentation[sh] OR methods[sh] OR ""Validation Studies""[pt] OR ""Comparative Study""[pt] OR ""psychometrics""[MeSH] OR psychometr*[tiab] OR clinimetr*[tw] OR clinometr*[tw] OR ""outcome assessment (health care)""[MeSH] OR ""outcome assessment""[tiab] OR ""outcome measure*""[tw] OR ""observer variation""[MeSH] OR ""observer variation""[tiab] OR ""Health Status Indicators""[Mesh] OR ""reproducibility of results""[MeSH] OR reproducib*[tiab] OR ""discriminant analysis""[MeSH] OR reliab*[tiab] OR unreliab*[tiab] OR valid*[tiab] OR ""coefficient of variation""[tiab] OR coefficient[tiab] OR homogeneity[tiab] OR homogeneous[tiab] OR ""internal consistency""[tiab] OR (cronbach*[tiab] AND (alpha[tiab] OR alphas[tiab])) OR (item[tiab] AND (correlation*[tiab] OR selection*[tiab] OR reduction*[tiab])) OR agreement[tw] OR precision[tw] OR imprecision[tw] OR ""precise values""[tw] OR test-retest[tiab] OR (test[tiab] AND retest[tiab]) OR (reliab*[tiab] AND (test[tiab] OR retest[tiab])) OR stability[tiab] OR interrater[tiab] OR inter-rater[tiab] OR intrarater[tiab] OR intra-rater[tiab] OR intertester[tiab] OR inter-tester[tiab] OR intratester[tiab] OR intra-tester[tiab] OR interobserver[tiab] OR inter-observer[tiab] OR intraobserver[tiab] OR intra-observer[tiab] OR intertechnician[tiab] OR inter-technician[tiab] OR intratechnician[tiab] OR intra-technician[tiab] OR interexaminer[tiab] OR inter-examiner[tiab] OR intraexaminer[tiab] OR intra-examiner[tiab] OR interassay[tiab] OR inter-assay[tiab] OR intraassay[tiab] OR intra-assay[tiab] OR interindividual[tiab] OR inter-individual[tiab] OR intraindividual[tiab] OR intra-individual[tiab] OR interparticipant[tiab] OR inter-participant[tiab] OR intraparticipant[tiab] OR intra-participant[tiab] OR kappa[tiab] OR kappa's[tiab] OR kappas[tiab] OR repeatab*[tw] OR ((replicab*[tw] OR repeated[tw]) AND (measure[tw] OR measures[tw] OR findings[tw] OR result[tw] OR results[tw] OR test[tw] OR tests[tw])) OR generaliza*[tiab] OR generalisa*[tiab] OR concordance[tiab] OR (intraclass[tiab] AND correlation*[tiab]) OR discriminative[tiab] OR ""known group""[tiab] OR ""factor analysis""[tiab] OR ""factor analyses""[tiab] OR ""factor structure""[tiab] OR ""factor structures""[tiab] OR dimension*[tiab] OR subscale*[tiab] OR (multitrait[tiab] AND scaling[tiab] AND (analysis[tiab] OR analyses[tiab])) OR ""item discriminant""[tiab] OR ""interscale correlation*""[tiab] OR error[tiab] OR errors[tiab] OR ""individual variability""[tiab] OR ""interval variability""[tiab] OR ""rate variability""[tiab] OR (variability[tiab] AND (analysis[tiab] OR values[tiab])) OR (uncertainty[tiab] AND (measurement[tiab] OR measuring[tiab])) OR ""standard | 9,508,385 |

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| | error of measurement""[tiab] OR sensitiv*[tiab] OR responsive*[tiab] OR (limit[tiab] AND detection[tiab]) OR ""minimal detectable concentration""[tiab] OR interpretab*[tiab] OR ((minimal[tiab] OR minimally[tiab] OR clinical[tiab] OR clinically[tiab]) AND (important[tiab] OR significant[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR (small*[tiab] AND (real[tiab] OR detectable[tiab]) AND (change[tiab] OR difference[tiab])) OR ""meaningful change""[tiab] OR ""ceiling effect""[tiab] OR ""floor effect""[tiab] OR ""Item response model""[tiab] OR IRT[tiab] OR Rasch[tiab] OR ""Differential item functioning""[tiab] OR DIF[tiab] OR ""computer adaptive testing""[tiab] OR ""item bank""[tiab] OR ""cross-cultural equivalence""[tiab])",Publication Date,,""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra-observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter-technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR ((""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR | |

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| | (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract]" | |
| #6 | "(""addresses""[Publication Type] OR ""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""lectures""[Publication Type] OR ""legal cases""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""popular works""[Publication Type] OR ""congresses""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms])",Publication Date,,"(""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms])" | 4,188,714 |
| #7 | #1 AND #2 AND #3 AND #4 AND #5,Publication Date,,"(""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]) AND (""Aged""[MeSH Terms] OR ""aged, 80 and over""[MeSH Terms] OR ""Frail Elderly""[MeSH Terms] OR ""Geriatrics""[MeSH Terms] OR ""Geriatric Psychiatry""[MeSH Terms] OR ""Geriatric Nursing""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Health Services for the Aged""[MeSH Terms] OR (""elder*""[Text Word] OR ""eldest""[Text Word] | 423 |

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| | OR ""frail*""[Text Word] OR ""geriatri*""[Text Word] OR ""old age*""[Text Word] OR ""oldest old*""[Text Word] OR ""senior*""[Text Word] OR ""senium""[Text Word] OR ""very old*""[Text Word] OR ""septuagenarian*""[Text Word] OR ""octagenarian*""[Text Word] OR ""octogenarian*""[Text Word] OR ""nonagenarian*""[Text Word] OR ""centarian*""[Text Word] OR ""centenarian*""[Text Word] OR ""supercentenarian*""[Text Word] OR ""older people""[Text Word] OR ""older subject*""[Text Word] OR ""older patient*""[Text Word] OR ""older age*""[Text Word] OR ""older adult*""[Text Word] OR ""older man""[Text Word] OR ""older men""[Text Word] OR ""older male*""[Text Word] OR ""older woman""[Text Word] OR ""older women""[Text Word] OR ""older female*""[Text Word] OR ""older population*""[Text Word] OR ""older person*""[Text Word])) AND (""nursing homes""[MeSH Terms] OR ""long term care""[MeSH Terms] OR ""residential care""[All Fields] OR ""homes for the aged""[MeSH Terms] OR ""residential facilities""[MeSH Terms] OR ""long-term care institution""[All Fields] OR ""residential aged care""[All Fields] OR ""skilled nursing facilities""[MeSH Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]) AND (""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]) AND (""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra- | |

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| | observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter-technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR (""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract])" | |
| #8 | #7 NOT #6,Publication Date,,"((""Oral Health""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Dentistry""[MeSH Terms] OR ""Dentists""[MeSH Terms] OR ""dental*""[Title] OR ""dentist*""[Title] OR ""oral hygien*""[Title] OR ""dental*""[Other Term] OR ""dentist*""[Other Term] OR ""oral hygien*""[Other Term]) AND (""Aged""[MeSH Terms] OR ""aged, 80 and over""[MeSH Terms] OR ""Frail Elderly""[MeSH Terms] OR ""Geriatrics""[MeSH Terms] OR ""Geriatric Psychiatry""[MeSH Terms] OR ""Geriatric Nursing""[MeSH Terms] OR ""Geriatric Dentistry""[MeSH Terms] OR ""Dental Care for Aged""[MeSH Terms] OR ""Health Services for the Aged""[MeSH Terms] OR (""elder*""[Text Word] OR ""eldest""[Text Word] OR ""frail*""[Text Word] OR ""geriatri*""[Text Word] OR ""old age*""[Text | 420 |

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| | Word] OR ""oldest old*""[Text Word] OR ""senior*""[Text Word] OR ""senium""[Text Word] OR ""very old*""[Text Word] OR ""septuagenarian*""[Text Word] OR ""octagenarian*""[Text Word] OR ""octogenarian*""[Text Word] OR ""nonagenarian*""[Text Word] OR ""centarian*""[Text Word] OR ""centenarian*""[Text Word] OR ""supercentenarian*""[Text Word] OR ""older people""[Text Word] OR ""older subject*""[Text Word] OR ""older patient*""[Text Word] OR ""older age*""[Text Word] OR ""older adult*""[Text Word] OR ""older man""[Text Word] OR ""older men""[Text Word] OR ""older male*""[Text Word] OR ""older woman""[Text Word] OR ""older women""[Text Word] OR ""older female*""[Text Word] OR ""older population*""[Text Word] OR ""older person*""[Text Word])) AND (""nursing homes""[MeSH Terms] OR ""long term care""[MeSH Terms] OR ""residential care""[All Fields] OR ""homes for the aged""[MeSH Terms] OR ""residential facilities""[MeSH Terms] OR ""long-term care institution""[All Fields] OR ""residential aged care""[All Fields] OR ""skilled nursing facilities""[MeSH Terms] OR ""nursing facility""[All Fields] OR ""care worker""[All Fields] OR ""nurses aids""[All Fields] OR ""nursing personnel""[All Fields] OR ""Caregivers""[MeSH Terms] OR ""Nursing""[MeSH Terms] OR ""Nursing""[MeSH Subheading] OR ""Nursing Care""[MeSH Terms] OR ""Nurses""[MeSH Terms] OR ""Nursing Staff""[MeSH Terms] OR ""Nursing Care""[MeSH Terms] OR ""Nurse's Role""[MeSH Terms] OR ""Home Health Aides""[MeSH Terms] OR ""Nursing Assistants""[MeSH Terms] OR ""Licensed Practical Nurses""[MeSH Terms] OR ""Home Care Services""[MeSH Terms] OR ""nurs*""[Title/Abstract] OR ""home care""[Title/Abstract] OR ""home healthcare""[Title/Abstract] OR ""home health care""[Title/Abstract] OR ""care aide*""[Title/Abstract] OR ""home health aide*""[Title/Abstract]) AND (""Oral Health screening tool""[All Fields] OR ""Geriatric Assessment""[MeSH Terms] OR ""assessment tool""[All Fields] OR ""needs assessment""[MeSH Terms] OR ""oral assessment""[All Fields] OR ""assessment scale""[All Fields] OR ""Surveys and Questionnaires""[MeSH Terms] OR ""questionnaire*""[Title/Abstract] OR ""instrument*""[Title/Abstract] OR ""measure*""[Title/Abstract] OR ""survey*""[Title/Abstract]) AND (""instrumentation""[MeSH Subheading] OR ""methods""[MeSH Subheading] OR ""Comparative Study""[Publication Type] OR ""psychometrics""[MeSH Terms] OR ""psychometr*""[Title/Abstract] OR ""clinimetr*""[Text Word] OR ""clinometr*""[Text Word] OR ""outcome assessment""[Title/Abstract] OR ""outcome measure*""[Text Word] OR ""observer variation""[MeSH Terms] OR ""observer variation""[Title/Abstract] OR ""Health Status Indicators""[MeSH Terms] OR ""reproducibility of results""[MeSH Terms] OR ""reproducib*""[Title/Abstract] OR ""discriminant analysis""[MeSH Terms] OR ""reliab*""[Title/Abstract] OR ""unreliab*""[Title/Abstract] OR ""valid*""[Title/Abstract] OR ""coefficient of variation""[Title/Abstract] OR ""coefficient""[Title/Abstract] OR ""homogeneity""[Title/Abstract] OR ""homogeneous""[Title/Abstract] OR ""internal consistency""[Title/Abstract] OR (""cronbach*""[Title/Abstract] AND (""alpha""[Title/Abstract] OR ""alphas""[Title/Abstract])) OR (""item""[Title/Abstract] AND (""correlation*""[Title/Abstract] OR ""selection*""[Title/Abstract] OR ""reduction*""[Title/Abstract])) OR ""agreement""[Text Word] OR ""precision""[Text Word] OR ""imprecision""[Text Word] OR ""precise values""[Text Word] OR ""test-retest""[Title/Abstract] OR (""test""[Title/Abstract] AND ""retest""[Title/Abstract]) OR (""reliab*""[Title/Abstract] AND (""test""[Title/Abstract] OR ""retest""[Title/Abstract])) OR ""stability""[Title/Abstract] OR ""interrater""[Title/Abstract] OR ""inter-rater""[Title/Abstract] OR ""intrarater""[Title/Abstract] OR ""intra-rater""[Title/Abstract] OR ""intertester""[Title/Abstract] OR ""inter-tester""[Title/Abstract] OR ""intratester""[Title/Abstract] OR ""intra-tester""[Title/Abstract] OR ""interobserver""[Title/Abstract] OR ""inter-observer""[Title/Abstract] OR ""intraobserver""[Title/Abstract] OR ""intra-observer""[Title/Abstract] OR ""intertechnician""[Title/Abstract] OR ""inter- | |

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| | technician""[Title/Abstract] OR ""intratechnician""[Title/Abstract] OR ""intra-technician""[Title/Abstract] OR ""interexaminer""[Title/Abstract] OR ""inter-examiner""[Title/Abstract] OR ""intraexaminer""[Title/Abstract] OR ""intra-examiner""[Title/Abstract] OR ""interassay""[Title/Abstract] OR ""inter-assay""[Title/Abstract] OR ""intraassay""[Title/Abstract] OR ""intra-assay""[Title/Abstract] OR ""interindividual""[Title/Abstract] OR ""inter-individual""[Title/Abstract] OR ""intraindividual""[Title/Abstract] OR ""intra-individual""[Title/Abstract] OR ""interparticipant""[Title/Abstract] OR ""inter-participant""[Title/Abstract] OR ""intraparticipant""[Title/Abstract] OR ""intra-participant""[Title/Abstract] OR ""kappa""[Title/Abstract] OR ""kappa's""[Title/Abstract] OR ""kappas""[Title/Abstract] OR ""repeatab*""[Text Word] OR (""replicab*""[Text Word] OR ""repeated""[Text Word]) AND (""measure""[Text Word] OR ""measures""[Text Word] OR ""findings""[Text Word] OR ""result""[Text Word] OR ""results""[Text Word] OR ""test""[Text Word] OR ""tests""[Text Word])) OR ""generaliza*""[Title/Abstract] OR ""generalisa*""[Title/Abstract] OR ""concordance""[Title/Abstract] OR (""intraclass""[Title/Abstract] AND ""correlation*""[Title/Abstract]) OR ""discriminative""[Title/Abstract] OR ""known group""[Title/Abstract] OR ""factor analysis""[Title/Abstract] OR ""factor analyses""[Title/Abstract] OR ""factor structure""[Title/Abstract] OR ""factor structures""[Title/Abstract] OR ""dimension*""[Title/Abstract] OR ""subscale*""[Title/Abstract] OR (""multitrait""[Title/Abstract] AND ""scaling""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""analyses""[Title/Abstract])) OR ""item discriminant""[Title/Abstract] OR ""interscale correlation*""[Title/Abstract] OR ""error""[Title/Abstract] OR ""errors""[Title/Abstract] OR ""individual variability""[Title/Abstract] OR ""interval variability""[Title/Abstract] OR ""rate variability""[Title/Abstract] OR (""variability""[Title/Abstract] AND (""analysis""[Title/Abstract] OR ""values""[Title/Abstract])) OR (""uncertainty""[Title/Abstract] AND (""measurement""[Title/Abstract] OR ""measuring""[Title/Abstract])) OR ""standard error of measurement""[Title/Abstract] OR ""sensitiv*""[Title/Abstract] OR ""responsive*""[Title/Abstract] OR (""limit""[Title/Abstract] AND ""detection""[Title/Abstract]) OR ""minimal detectable concentration""[Title/Abstract] OR ""interpretab*""[Title/Abstract] OR ((""minimal""[Title/Abstract] OR ""minimally""[Title/Abstract] OR ""clinical""[Title/Abstract] OR ""clinically""[Title/Abstract]) AND (""important""[Title/Abstract] OR ""significant""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR (""small*""[Title/Abstract] AND (""real""[Title/Abstract] OR ""detectable""[Title/Abstract]) AND (""change""[Title/Abstract] OR ""difference""[Title/Abstract])) OR ""meaningful change""[Title/Abstract] OR ""ceiling effect""[Title/Abstract] OR ""floor effect""[Title/Abstract] OR ""Item response model""[Title/Abstract] OR ""IRT""[Title/Abstract] OR ""Rasch""[Title/Abstract] OR ""Differential item functioning""[Title/Abstract] OR ""DIF""[Title/Abstract] OR ""computer adaptive testing""[Title/Abstract] OR ""item bank""[Title/Abstract] OR ""cross-cultural equivalence""[Title/Abstract])) NOT (""biography""[Publication Type] OR ""case reports""[Publication Type] OR ""comment""[Publication Type] OR ""directory""[Publication Type] OR ""editorial""[Publication Type] OR ""festschrift""[Publication Type] OR ""interview""[Publication Type] OR ""legislation""[Publication Type] OR ""letter""[Publication Type] OR ""news""[Publication Type] OR ""newspaper article""[Publication Type] OR ""patient education handout""[Publication Type] OR ""consensus development conference""[Publication Type] OR ""consensus development conference, nih""[Publication Type] OR ""practice guideline""[Publication Type]) NOT (""animals""[MeSH Terms] NOT ""humans""[MeSH Terms]))" | |

**PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol***

| Section and topic | Item No | Checklist item | Page Number |
|---|---|---|---|
| **ADMINISTRATIVE INFORMATION** | | | |
| Title: | | | |
| Identification | 1a | Identify the report as a protocol of a systematic review | 1 |
| Update | 1b | If the protocol is for an update of a previous systematic review, identify as such | NA |
| Registration | 2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | 2,5 |
| Authors: | | | |
| Contact | 3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | 1 |
| Contributions | 3b | Describe contributions of protocol authors and identify the guarantor of the review | 8 |
| Amendments | 4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | 4 |
| Support: | | | |
| Sources | 5a | Indicate sources of financial or other support for the review | 9 |
| Sponsor | 5b | Provide name for the review funder and/or sponsor | 9 |
| Role of sponsor or funder | 5c | Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol | 9 |
| **INTRODUCTION** | | | |
| Rationale | 6 | Describe the rationale for the review in the context of what is already known | 4 |
| Objectives | 7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | 4 |
| **METHODS** | | | |
| Eligibility criteria | 8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | 5 |
| Information sources | 9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | 5 |
| Search strategy | 10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be | Supplemental |

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| | | repeated | material |
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| Study records: | | | |
| Data management | 11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | 5,6 |
| Selection process | 11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | 5,6 |
| Data collection process | 11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | 6 |
| Data items | 12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | 6 |
| Outcomes and prioritization | 13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | 6 |
| Risk of bias in individual studies | 14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | 6-8 |
| Data synthesis | 15a | Describe criteria under which study data will be quantitatively synthesised | 7,8 |
| | 15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as $I^2$, Kendall's $\tau$) | 8 |
| | 15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | 8 |
| | 15d | If quantitative synthesis is not appropriate, describe the type of summary planned | 8 |
| Meta-bias(es) | 16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | 8 |
| Confidence in cumulative evidence | 17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | 8 |

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

*From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.*

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